



Cloud Computing

Assignment-Week 10

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

During live migration of a VM, which condition is most likely to increase the total migration time?

- A. Low rate of memory-page modification by the VM
- B. High rate of memory-page modification during migration
- C. Increasing the number of CPU cores on the destination host
- D. Reducing the VM memory size

Answer: (B)

Detailed Solution: In live migration, memory pages are copied while the VM continues running. If the VM modifies pages rapidly, many already-transferred pages become dirty and must be copied again. This increases the amount of data transferred and therefore increases migration time.

QUESTION 2:

Which live VM migration technique starts the VM on the destination first and then fetches memory pages from the source on demand as they are accessed?

- A) Cold migration
- B) Pre-copy migration
- C) Post-copy migration
- D) Offline image transfer

Answer: C

Detailed Solution: Post-copy migration resumes the VM on the destination immediately and then pulls memory pages from the source on demand (via page faults) or by background pushing. Pre-copy transfers memory before switching over, while cold and offline transfers require the VM to be stopped first.



QUESTION 3:

Which technique is most likely to reduce the total data transferred (and hence the energy used) during pre-copy live migration?

- A) Increasing the memory-dirtying rate of the workload
- B) Compressing and deduplicating memory pages before transfer
- C) Adding more iterative pre-copy rounds indefinitely
- D) Disabling dirty-page tracking

Answer: B

Detailed Solution: Memory compression and deduplication cut the volume of data sent over the network, reducing migration time and energy. A higher dirtying rate increases retransmissions, unlimited extra rounds waste bandwidth, and disabling dirty-page tracking breaks pre-copy correctness.

QUESTION 4:

Which resource management strategy is most important to prevent overload when many edge devices send data to nearby fog nodes?

- A) Increasing storage capacity at a single fog node
- B) Static task allocation
- C) Load balancing across fog nodes
- D) Sending all data directly to the cloud

Correct Answer: (C)

Detailed Solution: When many edge devices send data, one fog node can become overloaded. Static allocation or just adding storage doesn't handle changing workloads well. Load balancing spreads tasks across multiple fog nodes, avoiding congestion and keeping latency low. Sending all data to the cloud increases delay and bandwidth use, defeating the purpose of fog computing.

QUESTION 5:

A Kubernetes cluster has several worker nodes. A newly created Pod must be assigned to an appropriate node based on available resources and scheduling constraints.

Which Kubernetes component is primarily responsible for this decision?



- A. kubelet
- B. kube-scheduler
- C. kube-proxy
- D. Container registry

Correct Answer: B

Detailed Solution:

The **kube-scheduler** selects an appropriate worker node for newly created Pods. It considers factors such as resource requirements, node availability, affinity rules, and scheduling constraints. The kubelet subsequently manages the Pod on the selected node.

QUESTION 6:

Consider the following Docker workflow:

Dockerfile → docker build → _____ → docker run

Which entity is produced by the docker build operation?

- A. Docker volume
- B. Docker image
- C. Running container
- D. Kubernetes Pod

Correct Answer: B

Detailed Solution: The docker build command processes instructions in a Dockerfile and creates a Docker image. A container is subsequently created from an image when docker run is executed.

QUESTION 7:

Which of the following correctly describes the roles of Linux namespaces and cgroups in containers?

- A. Namespaces provide resource quotas, while cgroups provide process isolation.
- B. Namespaces provide isolation of processes and system resources, while cgroups control resource usage.



- C. Both namespaces and cgroups are used only to create Docker images.
- D. Neither is required for container isolation.

Correct Answer: B

Detailed Solution: Linux namespaces isolate processes and provide separate views of resources such as process IDs, networking, and mount points. cgroups restrict and monitor resource usage such as CPU and memory. Together, they provide important mechanisms for container isolation and resource control.

QUESTION 8:

A VM has 32 GB of memory and the migration link provides 16 MB/sec of usable throughput. For a non-live (cold) migration where the entire memory is transferred once, what is the approximate transfer time? (Use $1 \text{ GB} = 2^{30}$ bytes, $1 \text{ MB} = 2^{20}$ bytes.)

- A) About 15 minutes
- B) About 34 minutes
- C) About 1 hour
- D) About 2 hours

Answer: B

Solution: Time = $(32 \times 2^{30}) / (16 \times 2^{20}) = (32 \times 1024) / 16 = 32768 / 16 = 2048$ seconds ≈ 34 minutes.

QUESTION 9:

A database container is deleted accidentally. The database files were stored using a Docker named volume rather than only inside the container's writable layer. What happens to the data?

- A. The data is automatically deleted with the container.
- B. The data remains available in the named volume.
- C. The data is converted into a Docker image.
- D. The data can only be recovered by restarting the deleted container.



Correct Answer: B

Detailed Solution: Docker volumes have a lifecycle independent of individual containers. Deleting a container does not normally delete the data stored in its named volume. This makes volumes appropriate for persistent application data.

QUESTION 10:

Which of the following statements about containers and virtual machines is CORRECT?

- A. Containers normally require a separate complete guest operating system for every application.
- B. Virtual machines and containers always provide identical levels of isolation.
- C. Containers generally share the host operating system kernel, whereas VMs virtualize a complete machine environment.
- D. Containers cannot be deployed on virtual machines.

Correct Answer: C

Detailed Solution: Containers share the host OS kernel and therefore generally require fewer resources and start faster. A virtual machine provides a virtualized hardware environment in which a separate guest operating system runs. Containers can also run inside virtual machines, so option D is incorrect.